

A vision transformer-based deep neural network for state of health estimation of lithium-ion batteries

Liping Chen^{a,1}, Siqiang Xie^{a,1}, António M. Lopes^{b,*}, Xinyuan Bao^{a,1}

^a School of Electrical Engineering and Automation, Hefei University of Technology, Hefei, 230009, China

^b LAETA/INEGI, Faculty of Engineering, University of Porto, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal

ARTICLE INFO

Keywords:

Lithium-ion battery
State-of-health estimation
Transformer network

ABSTRACT

Accurate state-of-health (SOH) estimation of lithium-ion batteries (LIBs) is crucial to ensure the safety and reliability of electric vehicles. Deep learning has become a popular method for SOH estimation. However, this has mostly relied on convolutional neural networks (CNNs) and recurrent neural networks (RNNs), not fully exploring the potentialities of the method. This paper proposes a new procedure for SOH estimation of LIBs based on vision transformer networks (ViTs). By analyzing the training speed and accuracy for different sampling points, an adaptive algorithm is designed to choose the most appropriate sampled data for the ViT, guiding battery data collection in actual systems, and reducing manual work during neural network training. Moreover, the ViT is improved by adding a dimension transformation layer, a multilayer perceptron (MLP) and a trainable regression token. Experiments carried out on two different datasets revealed that the proposed framework is able to reach an accuracy better than 0.01, which is superior to that achieved with other available techniques. The new approach has high robustness, good accuracy and applicability, and the ViT has great potential in SOH estimation.

1. Introduction

With the deepening of people's attention to the issue of environmental protection, fossil energy is being replaced by renewable energy to power electric vehicles [1]. In new energy electric vehicles, the battery pack is one of the most important components. Lithium-ion batteries (LIBs) are widely used in the assembly of battery packs [2]. Compared with traditional secondary batteries, LIBs possess higher power density ratio and longer service life [3]. Besides vehicles, LIBs are also widely used in energy distribution and energy storage in power plants [4], being of great significance for reducing oil consumption and greenhouse gas emissions. Although LIBs have many advantages, they also have some drawbacks, such as susceptibility to external influences and accelerated aging [5]. Aged batteries are more prone to safety problems owing to their internal physical changes [6]. In electric vehicles, an aged battery not only decreases autonomy and performance, but also compromises safety, potentially leading to traffic accidents, fires, and other issues. Therefore, to maintain high efficiency and safe operation of electric vehicles, the state of health (SOH) of LIBs needs to be accurately determined [7]. However, the SOH cannot be measured directly by sensors, and the degradation of the LIB is a very complex process that depends on the working conditions. Therefore,

the SOH needs to be estimated from measurable variables, such as the battery voltage, current and temperature, being a very demanding process.

Several methods have been proposed to estimate the SOH of LIBs [8–10]. In general, they can be divided into (i) direct measurement, (ii) model-based, and (iii) data-driven methods. The direct measurement methods rely on the Coulomb counting [11], the open circuit voltage (OCV) [12], or the internal resistance estimation [13] approaches. Although the direct methods can be applied to any batteries, their main disadvantage is that they strongly depend on measurement technology. The model-based approaches use equivalent circuit [14] or electrochemical models [15]. The principle is simple and the computational burden is small, but usually lead to low estimation accuracy. Data-driven SOH prediction methods have gained increasing attention [16] because they do not require much computational effort to obtain relatively high-precision results. However, it is important to select appropriate data-driven algorithms. Support vector machines (SVMs) [17], Gaussian process regression (GPR) [18], and grey relationship analysis [19], have been adopted to learn nonlinear mappings between health features (HFs) and SOH. Additionally, sophisticated

* Corresponding author.

E-mail address: aml@fe.up.pt (A.M. Lopes).

¹ All authors contributed equally to this work.

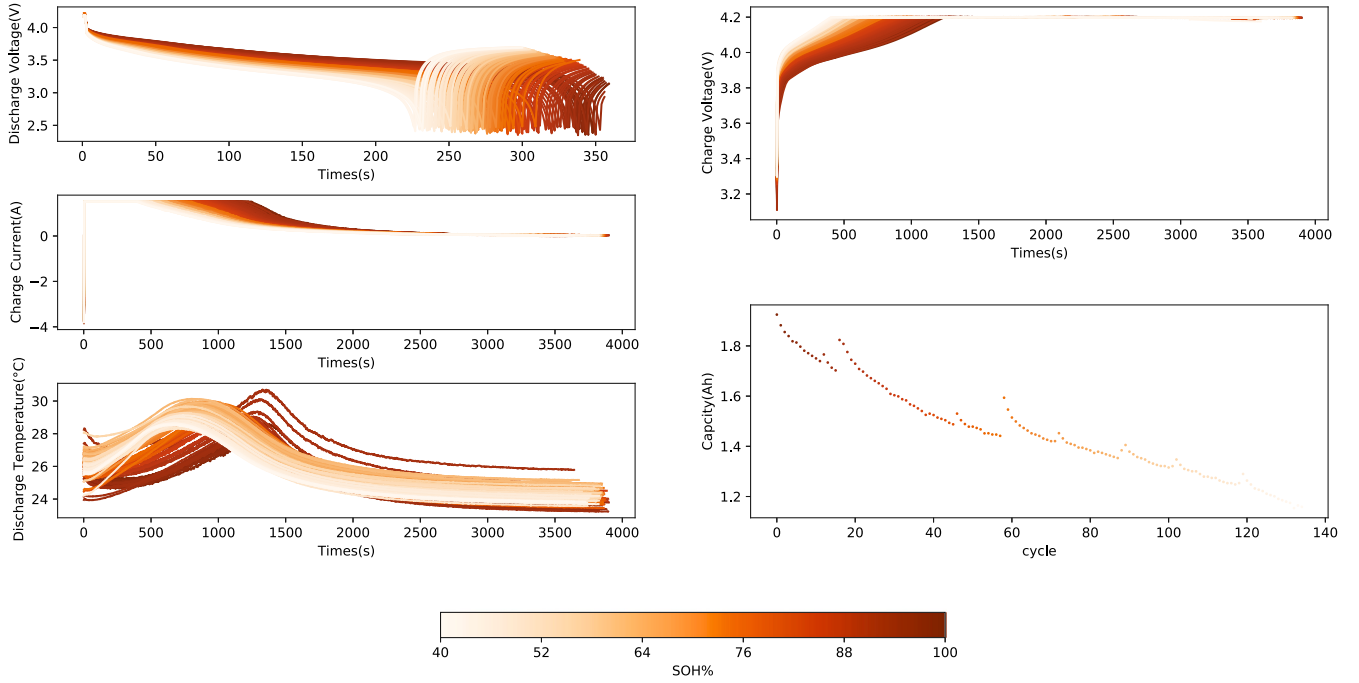


Fig. 1. The battery B0005 partial charge, discharge, temperature and capacity curves throughout the battery life cycle, according to the NASA dataset. Darker colors correspond to better SOH (fewer number of cycles).

machine learning methods, such as the unscented particle filter with optimized multiple kernel relevance vector machine (UPF-OMKRV) [20] has also been used to estimate SOH.

As a data-driven approach, the deep learning (DL) concept originated from recent developments in artificial neural networks and has been successfully applied in various fields. In the field of SOH estimation, various DL networks emerged, such as the CNN [21] and RNN [22]. In the meantime, researchers have made many improvements to these networks, such as the CNN combined with random forest [23], gated recurrent units (GRUs), long short-term memory (LSTM) neural networks [24], and hybrid CNN-GRU models [25]. These complex network structures and advanced optimization algorithms [26] can further improve the accuracy of SOH estimation. However, the existing optimization algorithms cannot be easily adopted in practical applications, and the contribution of complex networks to real applications is very limited due to long tuning time and slow convergence. Therefore, it is necessary to look for new neural networks with simple structure, and optimization algorithms with low computational cost.

The VIT network, based on the transformer network [27,28] has been successfully applied in the field of computer vision [29–31]. The VIT structure entirely abandons the traditional recurrence and convolution stages, and consists only of attention mechanisms. On the one hand, the VIT has a parallel structure, different from the traditional neural networks, which can improve the training speed. Moreover, the special attention mechanism can better complete tasks. However, the VIT network cannot be applied directly to SOH prediction, needing to be properly adapted.

To solve the above problems, this paper propose an algorithm that can select suitable sampling points, which can reduce the manual work in DL training. In addition, a regression vision transformer (RVIT) network is also established for SOH estimation of LIBs.

The contributions of the paper are as follows:

- (1) A novel VIT neural network is selected for SOH estimation, which can effectively capture the temporal and spatial information embedded in the battery data.
- (2) A RVIT network is established by improving the VIT in order to increase the accurately of SOH estimation. The RVIT can accommodate the multi-cycle and multi-sampling characteristics of the LIB data.

- (3) A new algorithm is proposed to reduce the manual work in network training by selecting the amount of battery sampling data.

The paper is organized as follows. Section 2 introduces the battery datasets used in the paper. Section 3 describes the algorithm proposed to adaptively select the sampled data to be used for SOH prediction. Section 4 presents the details of the new VIT-based model. Section 5 gives some illustrative examples of application. Finally, Section 6 summarizes the main conclusions.

2. Battery degradation datasets

2.1. Datasets source

In this paper, two types of datasets are adopted, namely those provided by NASA and University of Maryland. The NASA Battery Degradation Dataset was released by NASA's Ames Research Center. It contains degradation data for two sets of 18,650 LIBs with a nominal capacity of 2 Ah. The group corresponding to LIBs operation at room temperature (24 °C) was selected. The batteries numbered B0005, B0006, B0007 and B0018 are analyzed. The LIBs are repeatedly charged under constant current, constant voltage mode, from 1.5 A to 4.2 V, with a cutoff current of 20 mA. They are then discharged at a constant current of 2 A (CC) until their voltages drop to 2.7 V, 2.5 V, 2.2 V and 2.5 V, respectively. The University of Maryland Dataset is from the Center for Advanced Life Cycle Engineering (CALCE) [32]. The data are obtained under discharge rate of 1 C, discharge cutoff voltage of 2.7 V, charge rate of 0.5 C, and charge cutoff current of 50 mA, for all batteries. The charge and discharge rates remain constant, meaning that do not adjust with the attenuation of the battery capacity. The batteries numbered CS_35, CS_36, CS_37 and CS_38 are analyzed.

2.2. Datasets analysis

Fig. 1 depicts the battery B0005 partial charge, discharge, temperature and capacity curves, throughout the battery life cycle, according to the NASA dataset. Darker colors correspond to better SOH (fewer number of cycles). The color used in the capacity scatter plot referring to a specific number of cycles is the same adopted in the corresponding

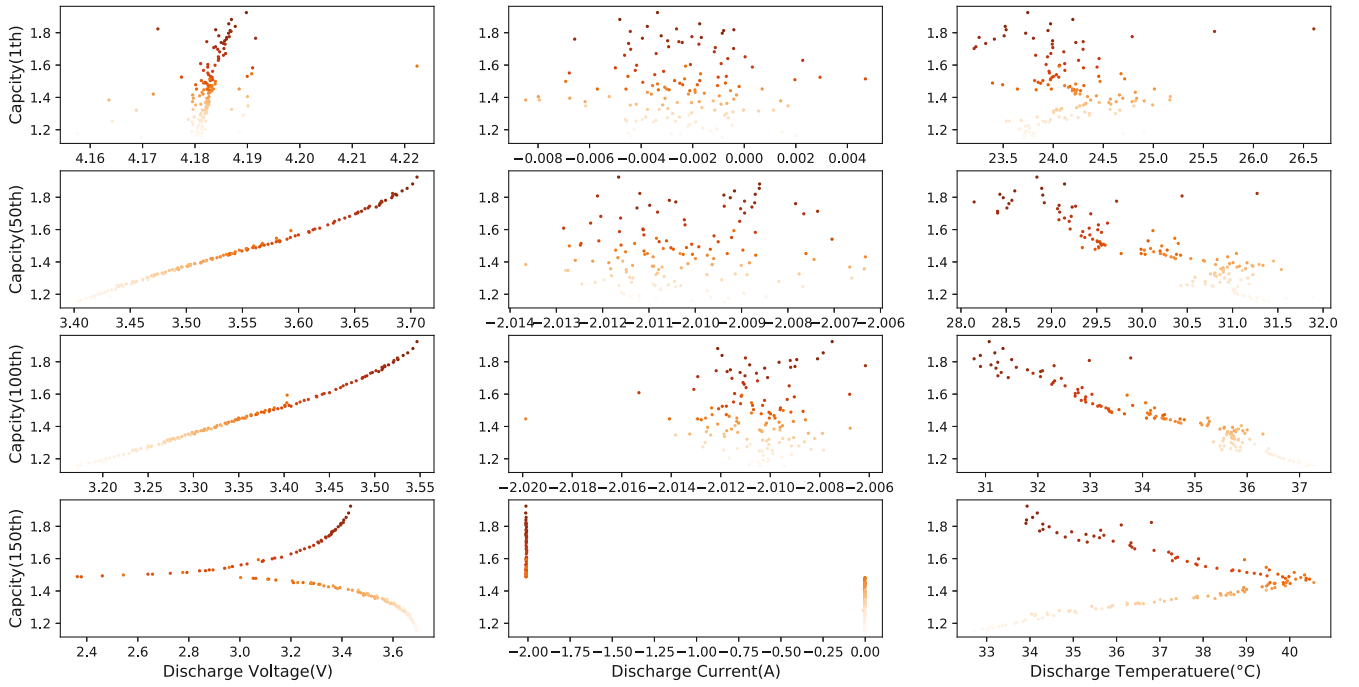


Fig. 2. The relationships between capacity and voltage, current and temperature, for the battery B0005, at four time instants.

charge, discharge and temperature curves. It can be verified that as the number of cycles increases, the time for the discharge (charge) voltage to reach its lowest (highest) value gets smaller. Similarly, the maximum temperature is reached faster. Moreover, the graphs show that parameters, such as current, voltage and temperature, change with the battery capacity during charging and discharging, meaning that these parameters can be used to estimate the battery capacity, which, in turn, can be used as an indicator of the SOH.

The SOH can be defined as:

$$SOH = \frac{C}{C_{norm}} \times 100\%, \quad (1)$$

where C and C_{norm} are the current and the nominal capacity of the battery, respectively.

3. Adaptive correlation analysis

Several studies only use some part of the sampled data to estimate SOH [25] and, recently, a one-to-one SOH estimation method based on resistance variation led to good results [33]. Fig. 2 depicts the relationships between capacity and voltage, current and temperature, for the battery B0005, at four different sampling time instants. Fig. 3 shows the Pearson correlation coefficients between HF's (current, voltage, temperature and their sum) and SOH at 30 different sampling times, given by the formula:

$$r = \frac{\sum_{i=1}^h (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^h (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^h (Y_i - \bar{Y})^2}}, \quad (2)$$

where h denotes the number of cycles, X represents the capacity over all cycles, Y corresponds to voltage, current or temperature at a given time instant, and $\bar{X}$ and $\bar{Y}$ represent average values. It can be seen that at some sampling times a poor correlation is obtained. Therefore, when training SOH prediction networks, it is not necessary to select all sampling points. On the contrary, if the number of sampling points can be reduced, the training phase will be faster.

This paper proposes an adaptive correlation analysis, as described by the algorithm flowchart depicted in Fig. 4. First, correlation is carried out, and the part of the data with higher correlation is selected.

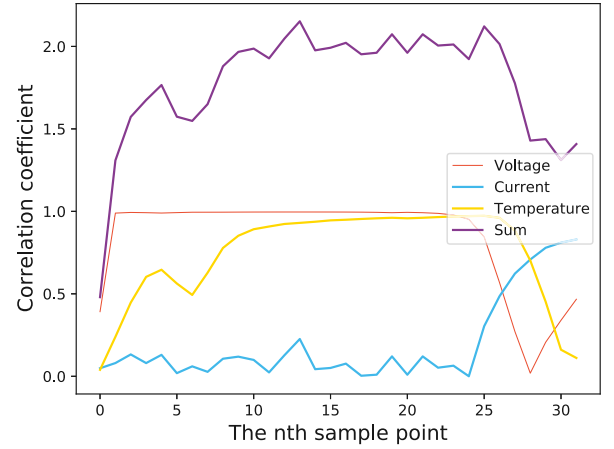


Fig. 3. The Pearson correlation coefficient at different sampling time instants.

At this stage, the number of sampling points per cycle is recorded as m . These m sampling points are then divided into n groups, each denoted by m_i , with the number of sampling points

$$m_{numi} = (m/n) \times i, \quad i = 1, 2, 3, \dots, n. \quad (3)$$

Due to the high correlation between these data and SOH, it is unnecessary to pay much attention to the order of the sampling points when grouping, as long as the order of all cycles is consistent.

In terms of speed analysis, taking the cases of batteries B0005, B0006, B0007 and B0018 as an example, as illustrated in Fig. 5, it can be seen that, for different number of sampling points, the loss function declines at different rates. Therefore, the training speed of each group is distinct and, for a given minimum loss function value, which can be determined by experience, the arrival time of different groups is different. The training completion time of group m_i is recorded and denoted by t_i .

In terms of precision analysis, the loss function of the regression problem can, to some extent, reflect the prediction accuracy, since both

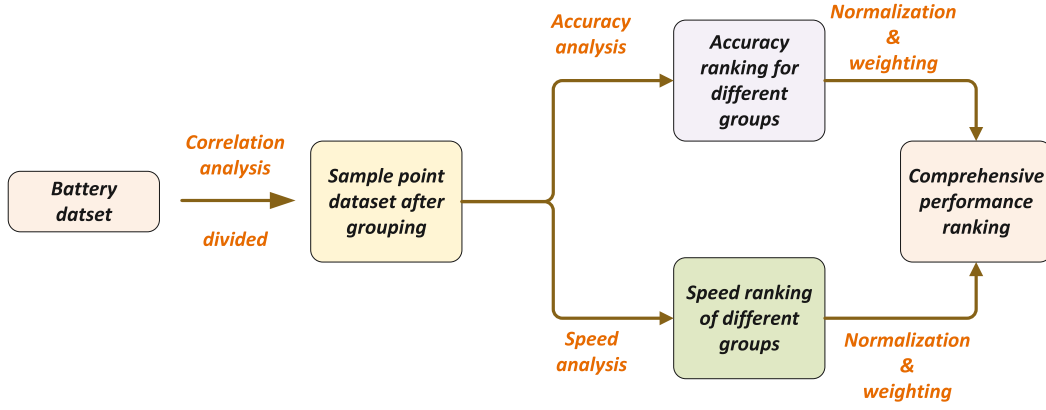


Fig. 4. The algorithm flowchart.

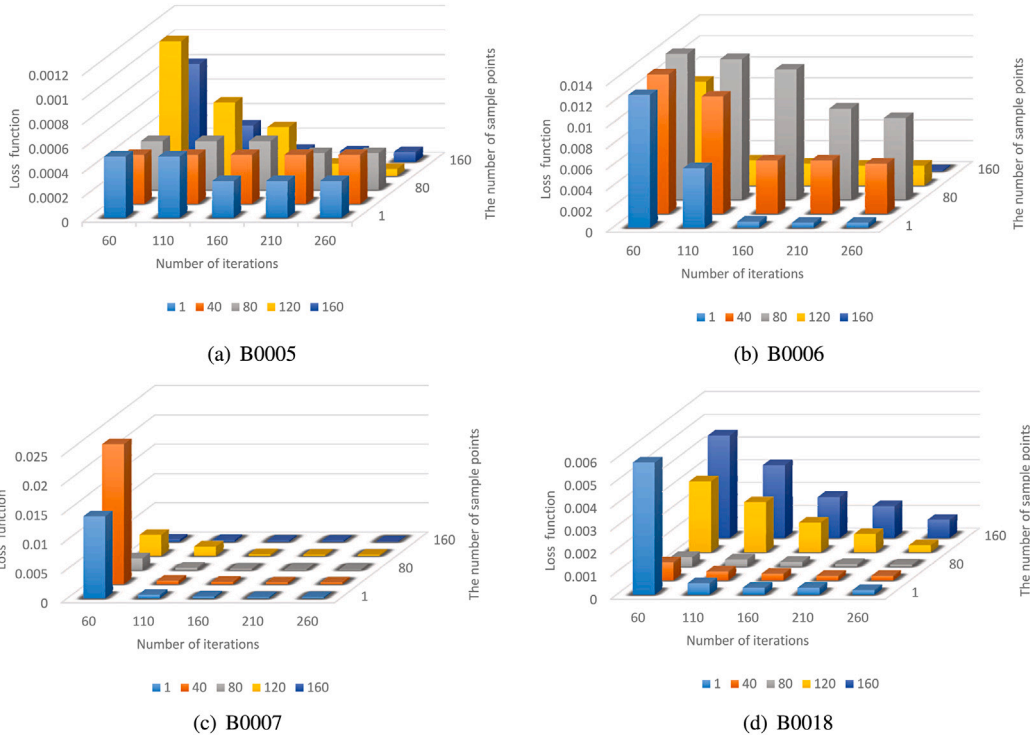


Fig. 5. The relationship between hyperparameters and loss functions of batteries B0005, B0006, B0007 and B018.

are trying to minimize the difference between the predicted and the actual values. Although a small loss function does not mean that the prediction error is small, if the loss function is high, then the prediction error is certainly high. Therefore, the loss function should be selected below the prediction error about three or four orders of magnitude, because a too small loss function can lead to overfitting. As such, when a reasonable loss function value is determined, it should have only three or four orders of magnitude worth predicting. A few loss functions for each order of magnitude should be selected and, then, the models corresponding to these loss functions are used to make predictions. The minimum prediction error of m_i is recorded and denoted by e_i . Notably, as the number of iterations of the SOH estimation network is very small, even all the smaller loss functions in the iteration process can be predicted, thus improving the accuracy of e_i .

After determining m_i , e_i and t_i , screening for e_i and t_i is carried out, since groups with too large e_i are not necessary for comparison. Given an error threshold value e_{norm} , it is determined $e_p = \{e_i : e_i <$

$e_{norm}, i \neq p\}$ and the corresponding t_p . Then, the values of e_p and t_p are normalized using the formula:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad (4)$$

where x and x' represent original and normalized data. Finally, a comprehensive performance evaluation criterion is obtained:

$$P_p = \alpha \times e_p' + \beta \times t_p', \quad (5)$$

where P_p is the performance evaluation index of group p , and α and β represent the weight of error and time, respectively, which can be adjusted by experience.

The data filtering can be completed automatically. By reducing the number of sampling points, the time for manually adjusting parameters can be reduced. Considering the influence of hyperparameters on the accuracy of the network, in the adaptive algorithm, the network model can be changed after adjusting the parameters for the first time, and

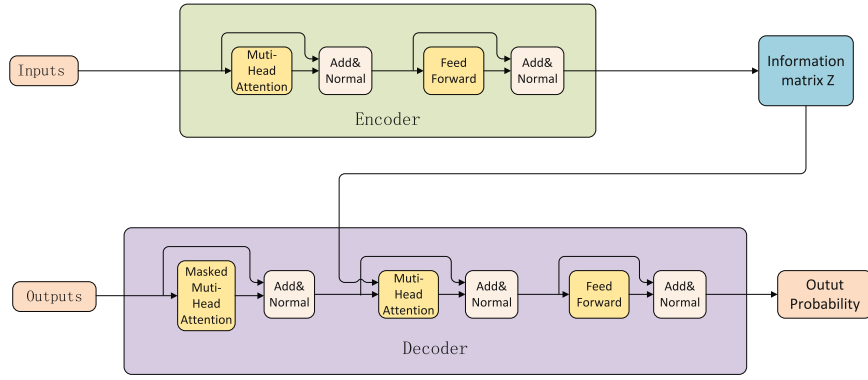


Fig. 6. Transformer structure.

then continue to select the optimal number of sample points. Considering that the number of samples described at the beginning of this section does not significantly improve the prediction accuracy, it is sufficient to select the faster group at one time.

4. Model description

4.1. Transformer network

Because the proposed RVIT network uses transformer encoder, the transformer is introduced first. Fig. 6 illustrates a schematic diagram of the transformer. Usually the information matrix Z is formed by encoding the input data. Then, Z is also used to complete the decoding task within the decoder.

The top half of Fig. 6 represents the transformer encoder process, while the bottom half shows the decoder. Multi-attention is composed of multiple self-attention blocks. Multi-attention is followed by simple normalization and feed-forward layers, for which explanation will be omitted herein. There can be multiple encoders and decoders in a network. Unlike conventional encoder and decoder structures, the transformer is entirely composed of attention mechanisms, called self-attention. Indeed, self-attention is at the heart of the transformer and is the key to its ability to achieve computing parallelism.

Fig. 7 shows the operation of the self-attention mechanism in practice. When a sequence $[x^1, x^2, x^3, x^4]$ is entered, after embedding, the matrix $[a^1, a^2, a^3, a^4]$ is obtained. The matrix elements then go into self-attention. They are multiplied by three transformation matrices, W^q , W^k and W^v , to get three different vectors, namely q (query), k (key) and v (value). That is, the vectors a^1, a^2, a^3 and a^4 have their own vectors q, k and v :

$$q^i = W^q a^i, \quad (6)$$

$$k^i = W^k a^i, \quad (7)$$

$$v^i = W^v a^i. \quad (8)$$

The method by which the a^i vector yields $\alpha_{1,i}$ is given in Eq. (9), where d_k represents the dimensions of k and q . Using the calculated q^i and k^i vectors of a^i , it is possible to relate sequences x^i entered at the same time. After getting $\alpha_{1,i}$, the softmax operation of Eq. (10) is performed on $\alpha_{1,i}$ to get $\bar{\alpha}_{1,i}$. But, since each input vector does not exactly act the same for the whole, the $\bar{\alpha}_{1,i}$ is weighted using the v^i and end up with the vector b^1 , as given in Eq. (11). As can be seen from the above operations, b^1 is a vector composed of x^1 and the information entered at the same time by x^2, x^3 and x^4 . Similarly, b^2, b^3 and b^4 are obtained, which is the principle of self-attention.

Eqs. (12) and (13) can be used to represent the encoder operation when entering matrix data, which stand for the operation of attention and the propagation of the fully connected layer, respectively. In

Eq. (12), Q, K, V represent q, k, v in matrix form and Eq. (13) is a simple fully connected layer formula. Fig. 8 shows the process by which the encoder portion of the transformer computes the information matrix Z when the input is a matrix, and the principle is consistent with the single input.

$$\alpha_{1,i} = q^i \cdot k^i / \sqrt{d_k}, \quad (9)$$

$$\bar{\alpha}_{1,i} = \exp(\alpha_{1,i}) / \sum_j \exp(\alpha_{1,j}), \quad (10)$$

$$b^1 = \sum_i \bar{\alpha}_{1,i} v^i, \quad (11)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (12)$$

$$\text{FFN}(Z) = \max(0, ZW_1 + b_1)W_2 + b_2. \quad (13)$$

So far, all components of the transformer have been introduced. It can be seen that for the same input sequence, attention is calculated in parallel, which is different from traditional RNNs. Indeed, the transformer calculates the input at all times simultaneously, while the RNN can only calculate sequentially. In [28], detailed theoretical and experimental proofs are given, which demonstrate the rapidity and accuracy of transformers. These features make it more effective than the RNNs in processing time series problems. Based on this, in the follow-up, a SOH estimation network is proposed.

4.2. Vision transformer

When directly applying the transformer to SOH estimation there are some issues. Indeed, the data for SOH are not fixed, and may consist of long full records and short parts sampled at specific time instants, which pose difficulties when encoding and decoding. The end-to-end structure increases the depth of the network. So, in the follow-up, it is adopted a VIT structure [27], which uses only the encoder part of the transformer. When the VIT is applied to SOH estimation, in addition to capturing the time-related information in HFs, the VIT can capture the information between different HFs through the “patch” section. In order to enable the VIT to be applied to state estimation of batteries, a dimension transformation layer is added at the front of the original network, a regression token to the VIT input layer is included to collect information related to SOH, and a regression prediction layer is joined at the end of the network, which together lead to the RVIT model.

Fig. 9 shows the steps for SOH estimation based on the RVIT. First, it is performed a dimensional transformation of the HFs, such as current, voltage and temperature, to convert one-dimensional into three-dimensional data. The two cubes in Fig. 9 illustrate two forms of dimensional transformation. The cube dimensions are length (L), width (W), and height (H). The upper cube can do many-to-one SOH

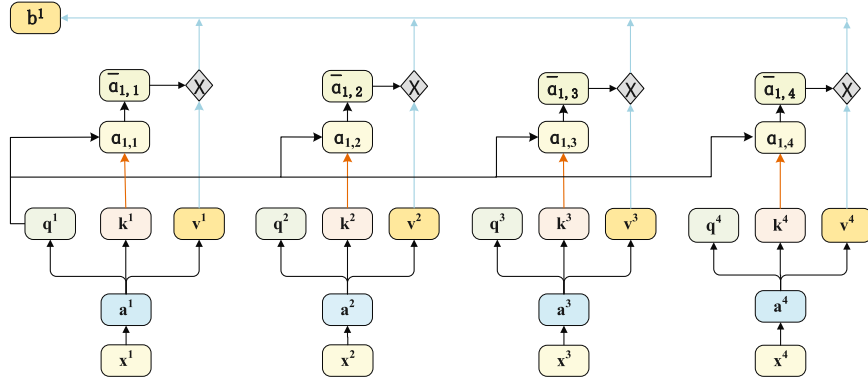


Fig. 7. Self-attention operation.

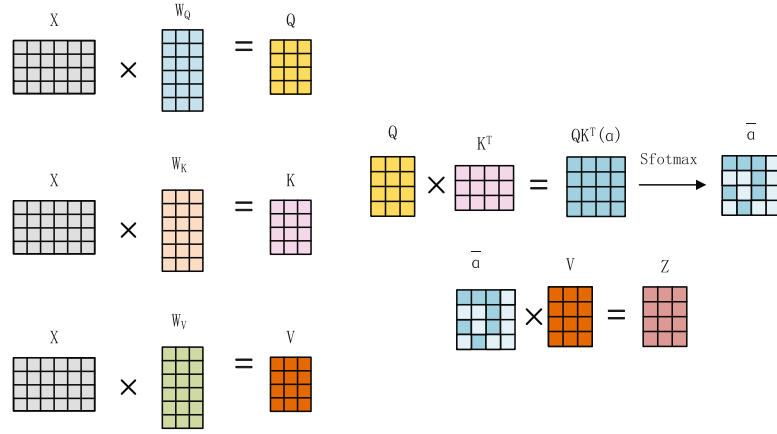


Fig. 8. The encoding action when data is entered as a matrix.

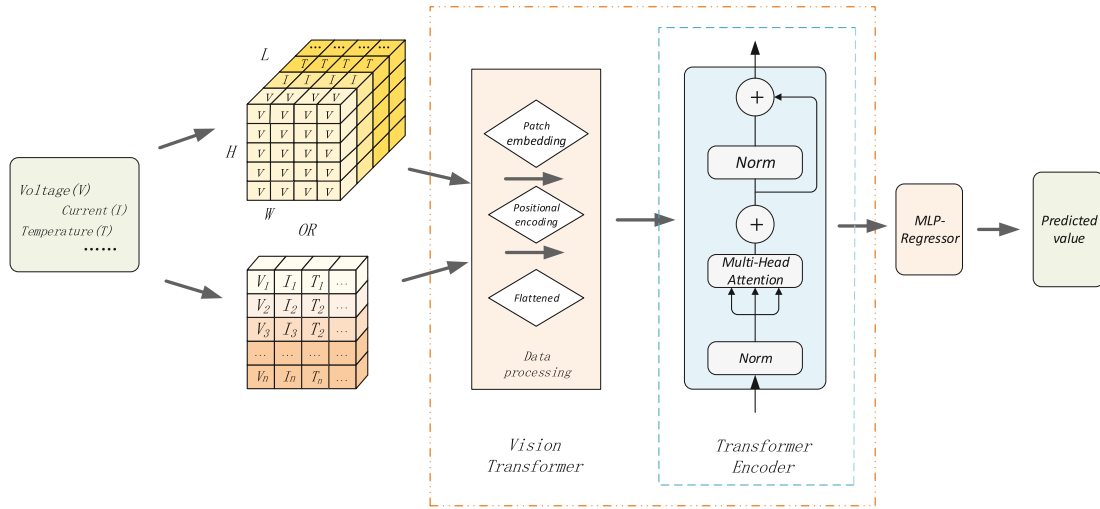


Fig. 9. The structure of VIT for SOH estimation.

estimation, where L represents the category of HF, and $H * W$ represents the total amount of a certain HF in all cycles. The bottom cube performs one-to-one SOH estimation, which is a special case when there is only one data value, for each parameter, per period. In this case, the dimension is also three-dimensional, but as $L = 1$, the cube dimensions are $H * W * 1$. Therefore, by adding a dimension transformation layer, the RVIT can perform many-to-one estimation and one-to-one estimation, without changing the structure of the network.

In the data preprocessing, the operation “patch embedding” will split the cube, because the original transformer can accept just limited size one-dimensional sequences. Assuming that the size of the blocks is $P_1 \times P_2 \times L$, a number N of blocks results, that is:

$$N = (H/P_1) \times (W/P_2). \quad (14)$$

Each piece, after dividing, is called x_i , with $i = 1, 2, 3, \dots, N$. Through the flattening operation, the matrix $[P_1, P_2, L]$ can be changed

to $[1, P_1 \times P_2 \times L]$, and the vector after that operation is called x_p^i , such that:

$$x_p^i = \text{flatten}(x_i), \quad i = 1, 2, 3, \dots, N. \quad (15)$$

Flattening is automatically computed, and x_p^i are designated as tokens. Because the original data are divided into N blocks, there are N tokens, and the total dimension is $[N, P_1 \times P_2 \times L]$. At this stage, battery sampling data are transformed into vectors that can be recognized by the transformer encoder. The input form of the transformer can be seen in [28]. Through different “patch” methods, the relationship between HFIs at different times can be captured, and the relationship between different kinds of HFIs can be apprehended. This ability is beyond those of the traditional RNN networks.

In the embedding operation, N tokens are obtained. In order to extract the information of all tokens, a trainable regression token x_r with dimension $[1, P_1 \times P_2 \times L]$ is added. Then, the regression token x_r is combined with other tokens x_p^i , and finally a token matrix with dimension $[N + 1, P_1 \times P_2 \times L]$ is obtained. Next, because the relative position of each vector in the patch is not taken into account, each token is assigned a position code. The dimension of the position code is consistent with that of the tokens, that is $[N + 1, P_1 \times P_2 \times L]$. The adopted position code is the one in reference [27]. Then, the two matrices are added, yielding a final matrix still with dimension $[N + 1, P_1 \times P_2 \times L]$, such that:

$$Z_0 = [x_r; x_p^1; x_p^2; \dots; x_p^N] + E_{POS}, \quad (16)$$

where Z_0 is the vector to be input into the transformer encoder at last, x_r is the trainable regression vector used for SOH prediction, x_p^i is the vector after flattening, used for extracting information and training x_r , and E_{POS} is the position information matrix with dimension $[N + 1, P_1 \times P_2 \times L]$, used to provide position information. Now, the matrix is sent to the transformer encoder for training, and the resulting x_r will pass through an MLP regression layer to predict the SOH. The transformer encoder as described above is a part of the transformer. The final capacity prediction formula is:

$$C_p = \text{MLP}(x_r'), \quad (17)$$

where C_p represents the final predicted capacity, MLP is a multi-layer perceptron capable of regression prediction, and x_r' is the vector output by the transformer encoder after x_r training.

5. Results and discussion

5.1. Experimental preparation

To verify the effectiveness of the proposed SOH prediction framework, the model is trained and tested with real battery data. The hardware configuration adopted for numerical processing consists of an Intel® Core™ i7-10700KF CPU @3.80 GHz, 32 GB memory, and NVIDIA GeForce GTX 3080ti graphics card. The model is implemented in Python 3.6 using the DL package ‘Torch’. Two public datasets, from NASA and University of Maryland, as described earlier in Section 2 are used. Electric vehicles batteries are taken as a good example, because most often users are unable to fully charge the battery, as well as to charge it to the same level. To estimate the SOH, data corresponding to the discharge interval are used. To better train the model, the dataset is uniformized by resampling, so that there are 660 sampling points per parameter and cycle. To mitigate the existence of outliers in the measured battery capacity, the data are cleaned up and the outliers replaced with the average of adjacent values.

After datasets preparation, the adaptive correlation procedure established in Section 3 is used, the appropriate parameters of sample data are selected, and SOH prediction is performed.

For the NASA datasets, the voltage, current, temperature, time, and capacity data are employed. For the CACLE, as the battery data are

measured at constant temperature, only voltage, current, and capacity are used. In both, the first 70% battery data are used to train, while the remaining 30% are used to test the model. The maximum absolute error (MAE) and the root mean squared error (RMSE) are adopted for assessing prediction accuracy:

$$MSE = \frac{1}{h} \sum_{i=1}^h (\hat{y}_i - y_i)^2, \quad (18)$$

$$RMSE = \sqrt{\frac{1}{h} \sum_{i=1}^h (\hat{y}_i - y_i)^2}, \quad (19)$$

where $\hat{y}$ represents the predicted value, y represents the true value, and h represents the total number of predictions.

In DL training, some hyperparameters are very important, and their setting affects the accuracy of the training [34]. The adjustment of the hyperparameters in CNNs and RNNs is very demanding [35,36]. Because the VIT completely abandons the traditional RNN and CNN structure, it is enough to set the number of encoder layers and the heads of self-attention. With the NASA dataset the batch is 20, the number of iterations is 200, the depth is 8, the number heads is 2, and the learning rate is 0.001. With the CALCE dataset the batch is set to 1024, the number of iterations is 200, the depth is 16, the number of heads is 2, and the learning rate is 0.0001. It can be seen that the difference between the two sets of hyperparameters is small, but both achieve good prediction results. This shows that the VIT has an absolute advantage in predicting SOH in terms of parameter adjustment. The loss function acts as a switch for the network to transfer information. Herein, it is adopted the GELU function, given by:

$$\text{GELU}(x) = 0.5x \left(1 + \tanh \left(\sqrt{2/\pi} (x + 0.044715x^3) \right) \right), \quad (20)$$

where x represents the input of neuron. The GELU function is the best loss function in the transformer network. In addition, in the proposed network structure, experiments demonstrated that the GELU can indeed get excellent precision. The results obtained with the proposed RVIT framework are compared with those achieved by traditional CNNs [37], RNNs [38], and GRU-CNNs [39]. In addition, because the data used for SOH forecasting are inconsistent, some additional comparisons considering recent SOH prediction methods were included. Thus, the RVIT framework is also compared with the LSTM based on empirical decomposition [40], incremental capacity analysis based on support vector regression [41], hybrid vector machine of UPF-OMKRVIM [20] and end-to-end SOH estimation structure [42].

5.2. Experimental results and analysis

Figs. 10 and 11 show the SOH prediction results obtained with NASA and CALCE datasets, for the batteries B005, B006, B007 and B018, and CS_35, CS_36, CS_37, and CS_38, respectively. Compared to the CNN or GRU alone, as can be seen, the GRU-CNN yields results closer to the true ones. This is because the two neural networks have different emphases when extracting data features. Indeed, the CNN focuses on extracting the relationships between different kinds of parameters, such as the relationship between voltage and current. The GRU, prefers the features of extracting sequence data, paying more attention to the relationship between the same parameter at different moments. The more extensive the feature extraction, the better the prediction accuracy, so the GRU-CNN shows better accuracy. But the disadvantage is that a larger model is required, which means that it takes longer to train and tune, so it may compromise online estimation. The RVIT yields better results than the others, surpassing the GRU-CNN. This is because the RVIT inherits the self-attention mechanism of the transformer, which enables the RVIT network to pay attention to the sequence relationship between the same kind of parameters, which allows it to completely replace the feature capture capability of the GRU. The structure of the ‘patch’ in the RVIT allows it to pay attention

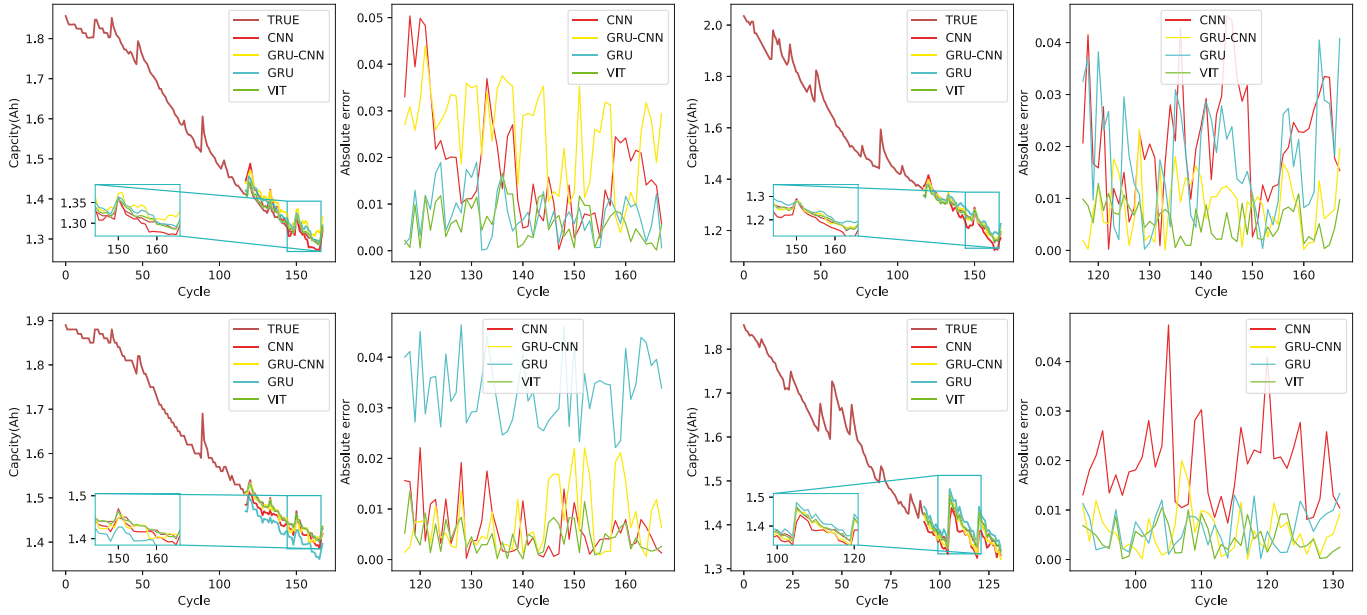


Fig. 10. The SOH prediction curves of NASA's datasets.

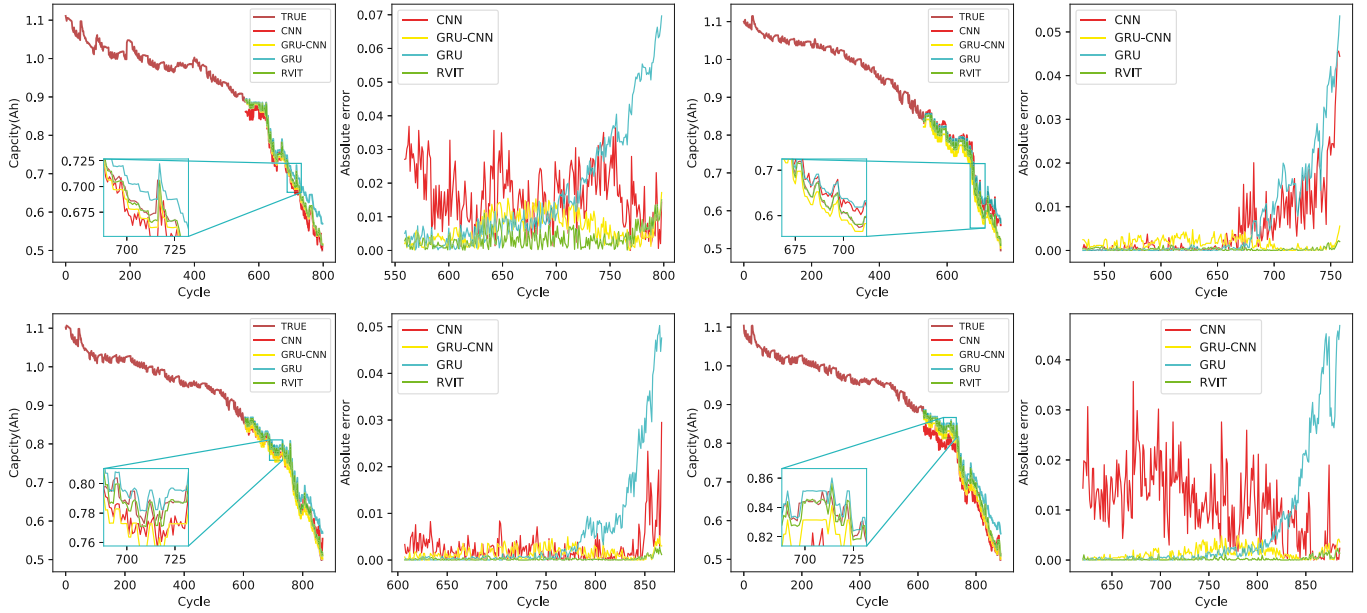


Fig. 11. The SOH prediction curves of CACLE's datasets.

to the relationship between different parameters, and also replaces the feature capture ability of the CNN to a certain extent. Due to its specific parallel computing mechanism, the RVIT training speed is far faster than that of the traditional RNN-derived networks, such as the GRU and LSTM. This property is also fully demonstrated in the CACLE datasets, where the RVIT also achieves very good results.

Tables 1 and 2 provide a clearer view of the estimation accuracy for the two datasets under different networks. It is noticed that the RVIT RMSE on the NASA dataset reached 0.46%, while the MAE is 0.36%. The RMSE on the CACLE dataset was 0.47% and the MAE was 0.37%. With the development of the battery industry, the capacity and power of the batteries will become larger. Due to its parallel structure, the transformer has a better predictive behavior on big data than other networks.

The prediction results based on the UPF-OMKRV [20] and regression support vector (ICA & SVR) [41] are also given in Tables 1 and 2, respectively. The prediction accuracy achieved by the LSTM based on empirical decomposition [40], with the CACLE dataset, is only 0.02, so it is not listed in the tables.

It should be noticed that the end-to-end SOH estimation structure is also an encoder–decoder structure [42]. However, this traditional encoder structure still maintains the traditional CNN and RNN structure, and its training speed is far lower than that of the transformer, which can be computed in parallel. Thanks to the parallel structure of the transformer, the RVIT also has great advantages in terms of training speed. The time to reach prediction was tested under the same conditions. The RVIT and CNN take both about 4 s, while the RNN takes about 6.5 s. When the data volume and the training effort increase, the

Table 1
Prediction error of NASA's battery dataset.

Battery number	Criteria (in %)	CNN	RNN	GRU-CNN [39]	UPF-OMKRVN [20]	RVIT
B0005	RMSE	2.15	0.92	2.71	0.51	0.43
	MAE	1.78	0.77	2.53	\	0.31
B0006	RMSE	2.1	1.8	0.88	0.60	0.51
	MAE	1.82	1.5	0.74	\	0.40
B0007	RMSE	0.67	3.42	0.97	0.41	0.39
	MAE	0.54	3.33	0.74	\	0.30
B0018	RMSE	2.08	0.73	0.71	1.09	0.54
	MAE	1.9	0.61	0.56	\	0.43
Average	RMSE	1.75	2.07	1.58	0.6525	0.46
	MAE	1.51	1.62	1.18	\	0.36

Table 2
Prediction error of CACLE's battery dataset.

Battery number	Criteria (in %)	CNN	RNN	GRU-CNN [39]	ICA & SVR [41]	RVIT
CS_35	RMSE	2.3	2.45	0.79	0.36	0.35
	MAE	1.65	1.71	0.68	0.27	0.25
CS_36	RMSE	2.28	2.58	0.89	0.67	0.55
	MAE	1.67	1.7	0.81	0.51	0.47
CS_37	RMSE	1.6	2.24	0.84	\	0.48
	MAE	1.35	1.66	0.75	\	0.37
CS_38	RMSE	3.34	2.53	0.78	\	0.5
	MAE	3.11	1.73	0.68	\	0.39
Average	RMSE	2.38	2.49	0.82	\	0.47
	MAE	1.94	1.7	0.73	\	0.37

RVIT only needs about half of the time of the RNN. Compared with the CNN, the RVIT can fully capture the information in time series, while this is difficult to complete for the CNN. The above results demonstrate the superior performance of the RVIT model, which proves the wide applicability of the RVIT.

6. Conclusion

A new method for SOH estimation of LIBs was proposed. Firstly, by considering both training speed and prediction accuracy, suitable sampling points were selected as the input data for a DL network, with the objective of reducing the time needed for manual training and parameter adjustment. Then, a RVIT network was established for SOH estimation, based on the VIT model, thus taking advantage of the VIT abilities for capturing information and parallel computing. Finally, two different datasets were adopted to test the effectiveness of the proposed method. The results showed that the RVIT achieved a fairly high level of accuracy, with a RMSE on the NASA and CACLE datasets reaching 0.46% and 0.47%, respectively. However, it should be noted that the RVIT may not be the optimal transformer structure for SOH prediction, and the VIT-based structure proposed in this paper may have still room for improvement. In future work, research on the potential of transformer structures, focusing on how these networks can yield better generalization capabilities and prediction accuracy for SOH, will be pursued.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Liping Chen reports financial support was provided by National Natural Science Foundation of China.

Data availability

Data are publicly available.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 62073114; 11971032).

References

- [1] Yao L, Fang Z, Xiao Y, Hou J, Fu Z. An intelligent fault diagnosis method for lithium battery systems based on grid search support vector machine. *Energy* 2021;214:118866.
- [2] Li X, Xu J, Hong J, Tian J, Tian Y. State of energy estimation for a series-connected lithium-ion battery pack based on an adaptive weighted strategy. *Energy* 2021;214:118858.
- [3] Liu K. State estimation for advanced battery management: key challenges and future trends. *Renew Sustain Energy Rev* 2019;114(OCT.):109334.1–109334.13.
- [4] Cano ZP, Banham D, Ye S, Hintennach A, Lu J, Fowler M, Chen Z. Batteries and fuel cells for emerging electric vehicle markets. *Nat Energy* 2018;3(4):279–89.
- [5] Landini S, O'Donovan T. Novel experimental approach for the characterisation of lithium-ion cells performance in isothermal conditions. *Energy* 2021;214:118965.
- [6] Suri G, Onori S. A control-oriented cycle-life model for hybrid electric vehicle lithium-ion batteries. *Energy* 2016;96:644–53.
- [7] Voronov S, Frisk E, Krysander M. Data-driven battery lifetime prediction and confidence estimation for heavy-duty trucks. *IEEE Trans Reliab* 2018;1–17.
- [8] Eddahech A, Briat O, Bertrand N, Deléage J-Y, Vinassa J-M. Behavior and state-of-health monitoring of li-ion batteries using impedance spectroscopy and recurrent neural networks. *Int J Electr Power Energy Syst* 2012;42(1):487–94.
- [9] Yang Q, Xu J, Li X, Xu D, Cao B. State-of-health estimation of lithium-ion battery based on fractional impedance model and interval capacity. *Int J Electr Power Energy Syst* 2020;119:105883.
- [10] Ge M-F, Liu Y, Jiang X, Liu J. A review on state of health estimations and remaining useful life prognostics of lithium-ion batteries. *Measurement* 2021;174:109057.
- [11] He W, Williard N, Chen C, Pecht M. State of charge estimation for li-ion batteries using neural network modeling and unscented Kalman filter-based error cancellation. *Int J Electr Power Energy Syst* 2014;62:783–91.
- [12] Chen L, Lü Z, Lin W, Li J, Pan H. A new state-of-health estimation method for lithium-ion batteries through the intrinsic relationship between ohmic internal resistance and capacity. *Measurement* 2018;116:586–95.
- [13] Galeotti M, Cinà L, Giammanco C, Cordiner S, Di Carlo A. Performance analysis and SOH (state of health) evaluation of lithium polymer batteries through electrochemical impedance spectroscopy. *Energy* 2015;89:678–86.
- [14] Wang S, Fernandez C, Yu C, Fan Y, Cao W, Stroe D-I. A novel charged state prediction method of the lithium ion battery packs based on the composite equivalent modeling and improved splice Kalman filtering algorithm. *J Power Sources* 2020;471:228450.
- [15] Lotfi N, Landers RG, Li J, Park J. Reduced-order electrochemical model-based SOC observer with output model uncertainty estimation. *IEEE Trans Control Syst Technol* 2016;25(4):1217–30.
- [16] Zhang W, Li X, Li X. Deep learning-based prognostic approach for lithium-ion batteries with adaptive time-series prediction and on-line validation. *Measurement* 2020;164:108052.
- [17] Tao X, Li Q, Guo W, Ren C, Li C, Liu R, Zou J. Self-adaptive cost weights-based support vector machine cost-sensitive ensemble for imbalanced data classification. *Inform Sci* 2019;487:31–56.
- [18] Tabatabaeipour M, Tzaferis K, McMillan R, Jackson W, Dobie G, Edwards RS, Trushkevych O, Gachagan A. Ultrasonic guided wave estimation of minimum remaining wall thickness using Gaussian process regression. *Mater Des* 2022;221:110990.
- [19] Yuan X, Liu Y, Bucknall R. Optimised MOPSO with the grey relationship analysis for the multi-criteria objective energy dispatch of a novel SOFC-solar hybrid CCHP residential system in the UK. *Energy Convers Manage* 2021;243(February):114406.
- [20] Sun X, Zhong K, Han M. A hybrid prognostic strategy with unscented particle filter and optimized multiple kernel relevance vector machine for lithium-ion battery. *Measurement* 2021;170:108679.
- [21] Wang S, Ren P, Takyi-Aninakwa P, Jin S, Fernandez C. A critical review of improved deep convolutional neural network for multi-timescale state prediction of lithium-ion batteries. *Energies* 2022;15(14):5053.
- [22] Wang S, Fan Y, Jin S, Takyi-Aninakwa P, Fernandez C. Improved anti-noise adaptive long short-term memory neural network modeling for the robust remaining useful life prediction of lithium-ion batteries. *Reliab Eng Syst Saf* 2023;230:108920.
- [23] Yang N, Song Z, Hofmann H, Sun J. Robust state of health estimation of lithium-ion batteries using convolutional neural network and random forest. *J Energy Storage* 2022;48:103857.
- [24] Wang S, Takyi-Aninakwa P, Jin S, Yu C, Fernandez C, Stroe D-I. An improved feedforward-long short-term memory modeling method for the whole-life-cycle state of charge prediction of lithium-ion batteries considering current-voltage-temperature variation. *Energy* 2022;254:124224.

- [25] Mei P, Li M, Zhang Q, Li G, et al. Prediction model of drinking water source quality with potential industrial-agricultural pollution based on CNN-GRU-Attention. *J Hydrol* 2022;127934.
- [26] Wu J, Fang L, Dong G, Lin M. State of health estimation of lithium-ion battery with improved radial basis function neural network. *Energy* 2023;262:125380.
- [27] Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, et al. An image is worth 16x16 words: Transformers for image recognition at scale. 2020, arXiv preprint arXiv: 2010.11929.
- [28] Ashish V, Noam S, et al. Attention is all you need. In: Conference and workshop on neural information processing systems. 2017.
- [29] Han K, Wang Y, Chen H, Chen X, Guo J, Liu Z, Tang Y, Xiao A, Xu C, Xu Y, et al. A survey on vision transformer. *IEEE Trans Pattern Anal Mach Intell* 2022;45(1):87–110.
- [30] Wang X, Zhang LL, Wang Y, Yang M. Towards efficient vision transformer inference: a first study of transformers on mobile devices. In: Proceedings of the 23rd annual international workshop on mobile computing systems and applications. 2022, p. 1–7.
- [31] Hong W, Lao J, Ren W, Wang J, Chen J, Chu W. Training object detectors from scratch: An empirical study in the era of vision transformer. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022, p. 4662–71.
- [32] Miao Q, Xie L, Cui H, Liang W, Pecht M. Remaining useful life prediction of lithium-ion battery with unscented particle filter technique. *Microelectron Reliab* 2013;53(6):805–10.
- [33] Chen L, Ding Y, Wang H, Wang Y, Liu B, Wu S, Li H, Pan H. Online estimating state of health of lithium-ion batteries using hierarchical extreme learning machine. *IEEE Trans Transp Electr* 2021;8(1):965–75.
- [34] Cheng G, Wang X, He Y. Remaining useful life and state of health prediction for lithium batteries based on empirical mode decomposition and a long and short memory neural network. *Energy* 2021;232:121022.
- [35] Raziani S, Azimbagirad M. Deep CNN hyperparameter optimization algorithms for sensor-based human activity recognition. *Neurosci Inform* 2022;2(3):100078.
- [36] Li W, Ng WW, Wang T, Pelillo M, Kwong S. HELP: An LSTM-based approach to hyperparameter exploration in neural network learning. *Neurocomputing* 2021;442:161–72.
- [37] Zhou D, Wang B. Battery health prognosis using improved temporal convolutional network modeling. *J Energy Storage* 2022;51:104480.
- [38] Zhang R, Chen T, Xiao F, Luo J. Bi-directional gated recurrent unit recurrent neural networks for failure prognosis of proton exchange membrane fuel cells. *Int J Hydrogen Energy* 2022.
- [39] Fan Y, Xiao F, Li C, Yang G, Tang X. A novel deep learning framework for state of health estimation of lithium-ion battery, *J Energy Storage* 32.
- [40] Cheng G, Wang X, He Y. Remaining useful life and state of health prediction for lithium batteries based on empirical mode decomposition and a long and short memory neural network. *Energy* 2021;232:121022.
- [41] Zhang Y, Liu Y, Wang J, Zhang T. State-of-health estimation for lithium-ion batteries by combining model-based incremental capacity analysis with support vector regression. *Energy* 2022;239:121986.
- [42] Gong Q, Wang P, Cheng Z. An encoder-decoder model based on deep learning for state of health estimation of lithium-ion battery. *J Energy Storage* 2022;46:103804.